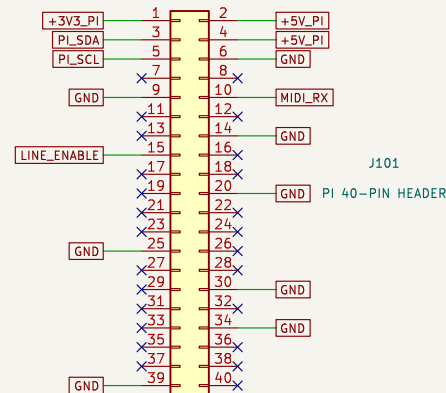


ARDOR / CODEC ZERO I/O

01 / System interface and power

REV C | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

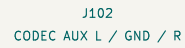
PI HEADER / STACKING INTERFACE



GPIO15 / pin 10: UART RX, 31250 baud, 8-N-1
GPIO17 / pin 11: unused by Rev C
GPIO27 / pin 13: reserved for Codec Zero button

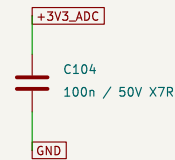
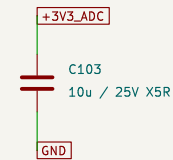
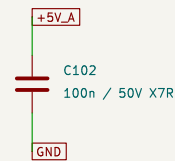
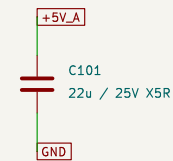
GPIO22 / pin 15: line relay enable, active high
I2C1: GPIO2/3. ADS1115 address 0x48.
Reserved Codec pins: GPIO18–21 I2S; GPIO23/24 LEDs;
GPIO27 button; GPIO0/1 HAT EEPROM. No added loads.

AUX HEADER HARNESS



J102 pin order is OUR harness definition.
Map to the actual Codec Zero L/R/GND pads.
Never connect the mono speaker BTL output here.

FILTERED BRANCHES / 5 V REGULATED INPUT ONLY



Audio jack sleeves bond GND to aluminium enclosure.
Use one continuous PCB ground plane; place by function.
R101 bonds TVS CHASSIS to GND; keep sleeve leads short.
5 V source: 4.75–5.25 V; reserve 100 mA beyond Pi/Codem.
Pi provides 3.3 V. No second supply or USB back-feed.



02 / Isolated MIDI input

File: 02_MIDI.kicad_sch

03 / Expression input

File: 03_Expression.kicad_sch

04 / Audio distribution

File: 04_Audio.kicad_sch

05 / Line and amp feed

File: 05_Line_Amp.kicad_sch

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /
File: Ardor_IO.kicad_sch

Title: 01 / System interface and power

Size: A3

Date: 2026-09-30

Rev: C

KiCad E.D.A. 9.0.2

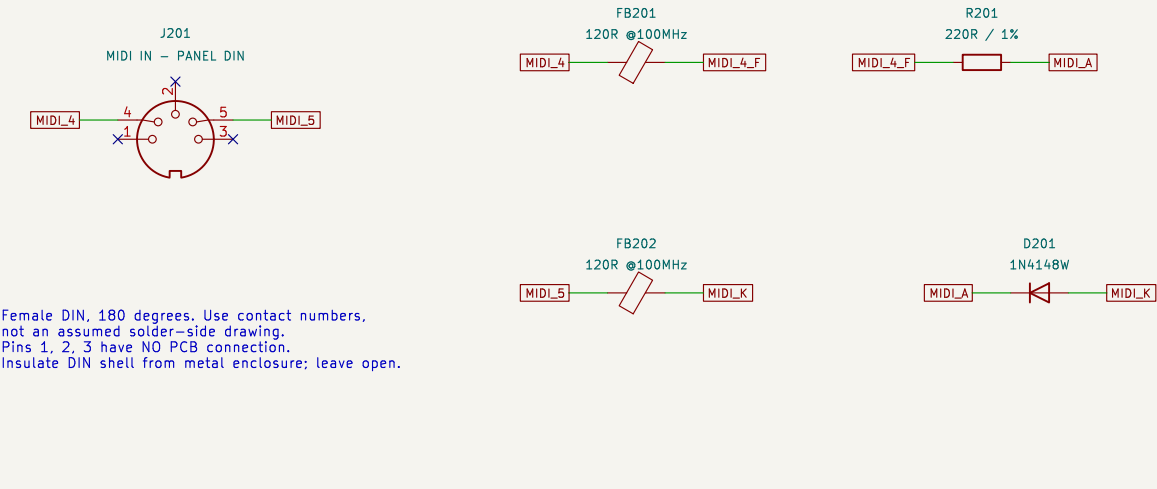
id: 1/5

ARDOR / CODEC ZERO I/O

02 / Isolated MIDI input

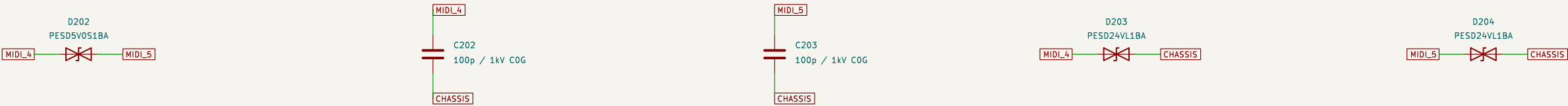
REV C | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

5–PIN DIN / FLOATING CURRENT LOOP



Female DIN, 180 degrees. Use contact numbers, not an assumed solder-side drawing. Pins 1, 2, 3 have NO PCB connection. Insulate DIN shell from metal enclosure; leave open.

CONNECTOR ESD / RF NETWORK



D202 clamps differential transients; D203/D204 divert common-mode ESD to CHASSIS (24 V standoff). Keep the MIDI loop copper isolated from logic GND: >= 3 mm clearance around U201 input and nets. Common-mode range is limited by D203/D204. Functional ground-loop isolation; NOT safety isolation. No MIDI-pin connection to logic supply. Place TVS and 100p RF capacitors at DIN harness entry. Qualify IEC 61000-4-2 on the assembled enclosure, including common-mode strikes and UART recovery.

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /02 / Isolated MIDI input/
File: 02_MIDI.kicad_sch

Title: 02 / Isolated MIDI input

Size: A3
KiCad E.D.A. 9.0.2

Date: 2026-09-30

Rev: C

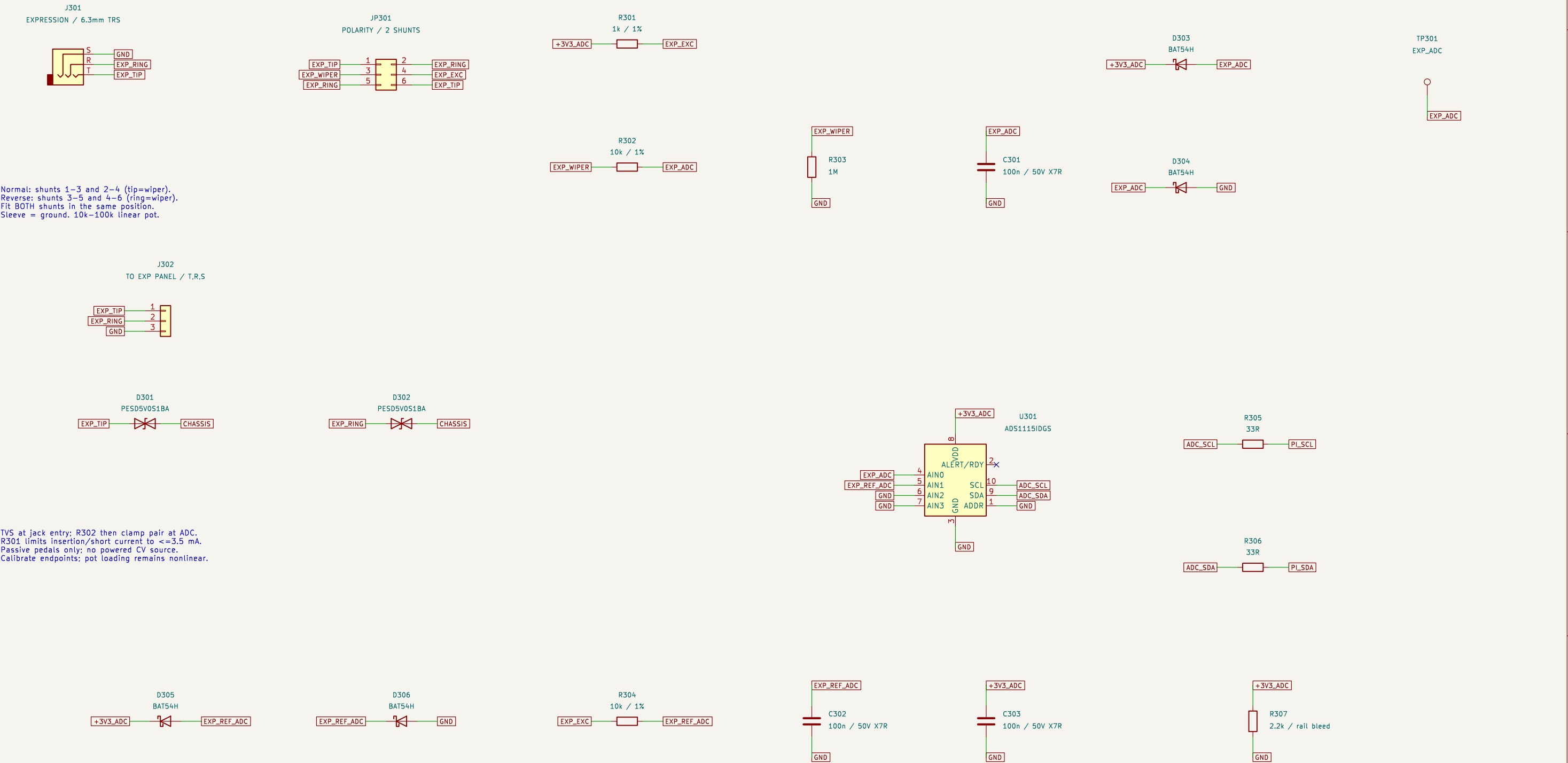
Id: 2/5

ARDOR / CODEC ZERO I/O

03 / Expression input

REV C | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

PASSIVE TRS PEDAL / TIP–RING SELECT



I2C address 0x48; internal reference; PGA +/-4.096 V; 128 SPS. Read AIN0 and AIN1 single-ended.
Use AIN0 / AIN1 ratio, endpoint calibration, smoothing and deadband. No added I2C pull-ups (Pi has them).

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /03 / Expression input/

File: 03_Expression.kicad_sch

Title: 03 / Expression input

Size: A3

Date: 2026-09-30

Rev: C

KiCad E.D.A. 9.0.2

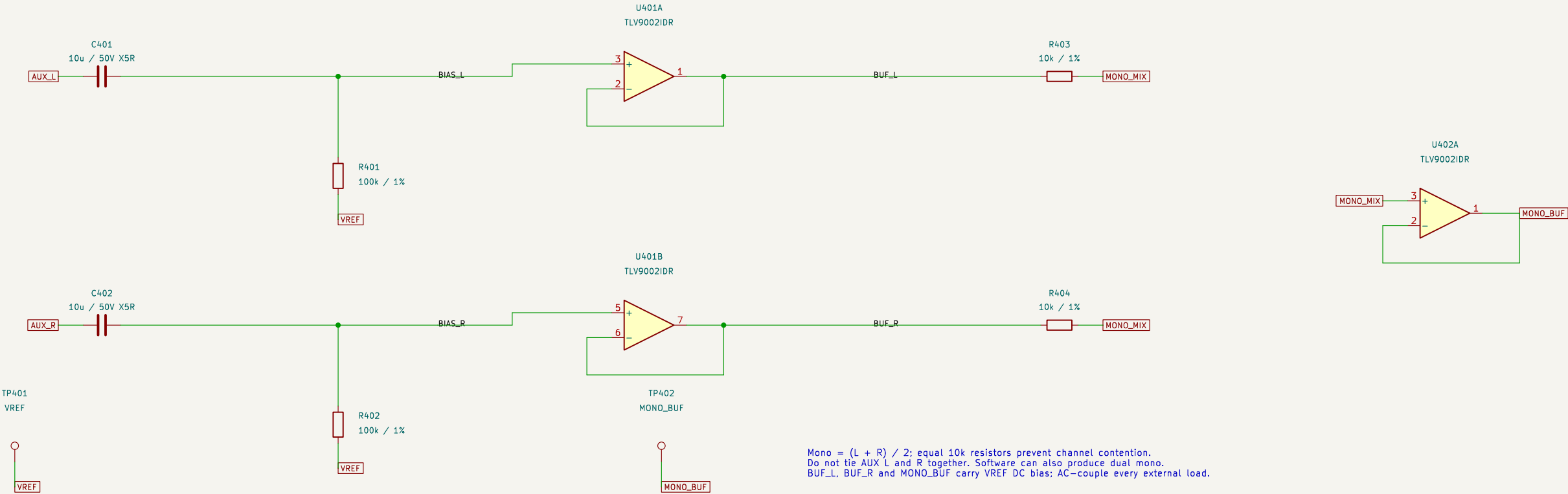
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ARDOR / CODEC ZERO I/O

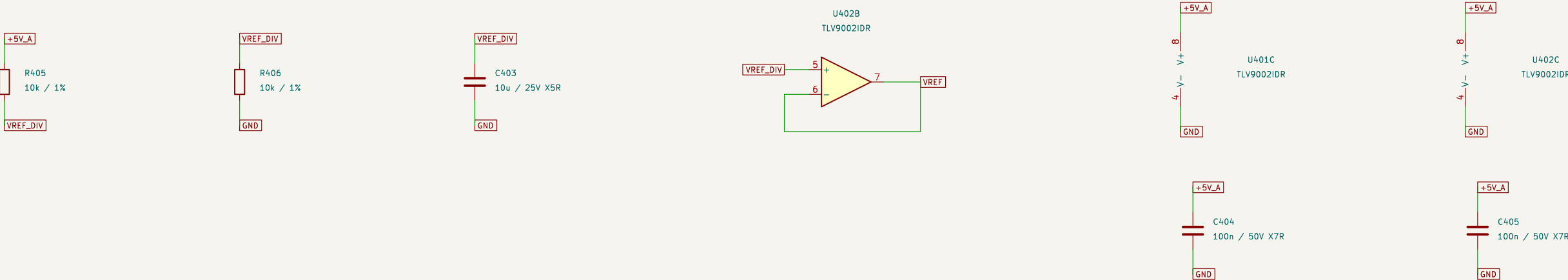
04 / Audio distribution

REV C | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

AC-COUPLED STEREO INPUT / UNITY BUFFERS



MIDRAIL / LOCAL BYPASS



Nominal audio target <=1.0 Vrms per AUX channel. Unity gain throughout.
TLV9002: RRIO on 5 V, unity stable; 2.5 V bias permits bipolar audio swing.
10u/100k input pole ~0.159 Hz. Keep VREF local; no large capacitor directly on buffer output.

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /04 / Audio distribution/
File: 04_Audio.kicad_sch

Title: 04 / Audio distribution

Size: A3 Date: 2026-09-30

KiCad E.D.A. 9.0.2

Rev: C

Id: 4/5

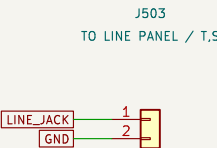
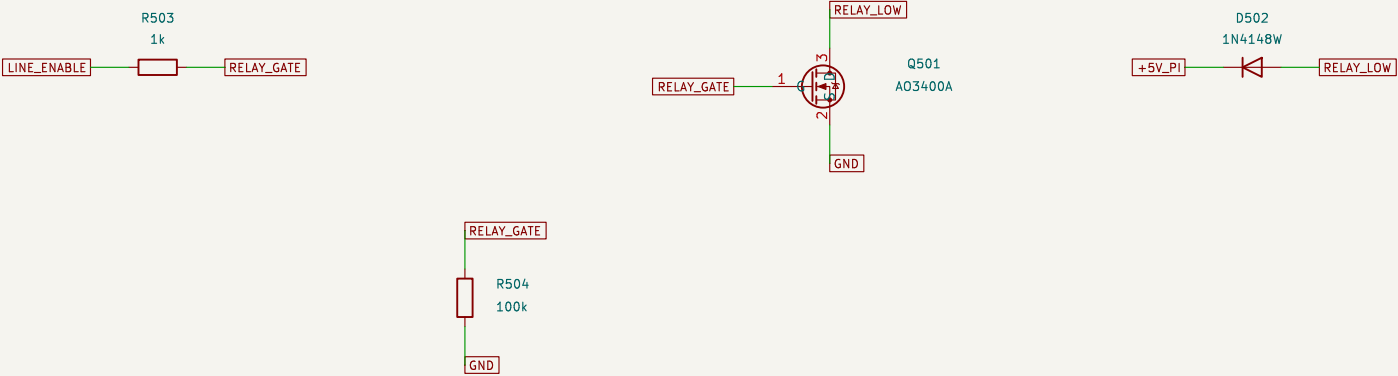
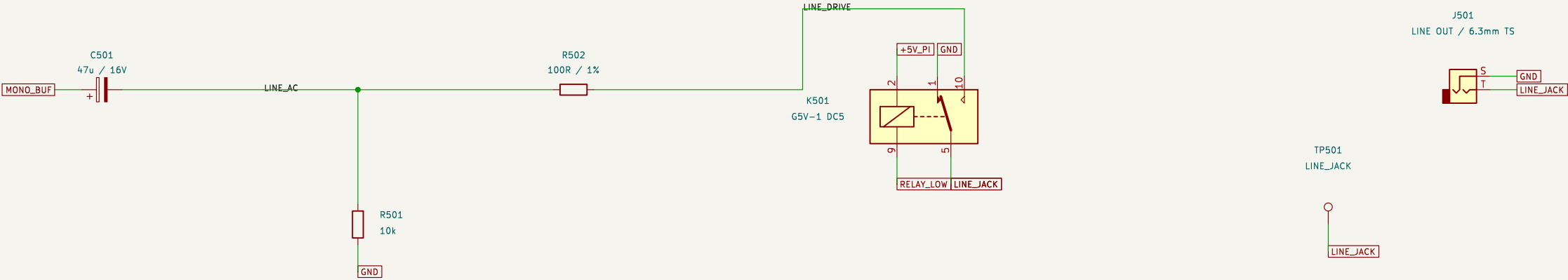
ARDOR / CODEC ZERO I/O

05 / Line and amp feed

REV C | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

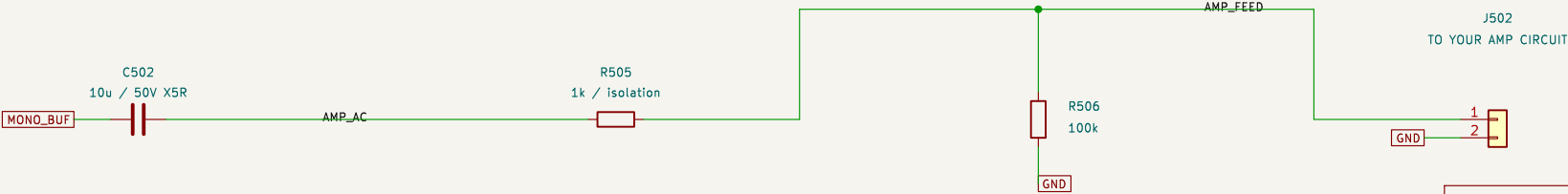
MONO LINE / DC BLOCK / STARTUP RELAY MUTE

Relay de-energized: jack tip grounded, source disconnected.
Energized: buffered mono reaches tip. Load >=10k.
C501 positive faces MONO_BUF; resistor outside feedback.
No balanced / +4 dBu / phantom-power capability specified.



Amp input >=100k. Add your gain, output connector,
external-port protection and mute in that project.
J502 is internal, NOT a speaker / amp output jack.

AMP CIRCUIT PLACEHOLDER / INTERNAL HARNESS



Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /05 / Line and amp feed/
File: 05_Line_Amp.kicad_sch

Title: 05 / Line and amp feed

Size: A3 Date: 2026-09-30
KiCad E.D.A. 9.0.2

Rev: C
Id: 5/5